Lean Six Sigma Practice Test

1. What is the key benefit of multi-vari charting?

A. It keeps track of the time when measurements were made
B. It graphically displays the variation in a process
C. It assists in the breakdown of components of variation
D. It is much easier to plot than most control charts

2. Deciding to collect a sample from a population is often:

A. A Personal decision
B. An Economical decision
C. A Risky decision
D. A Team decision

3. A Six Sigma project had a lot of improvement measures suggested on training people, setting standardized job instructions, getting surveys done from employees, and so on. Who would be the focused audience for whom this project was aimed at?

A. External customers
B. Stakeholders
C. Internal customers
D. All of the above

4. Cause and Effect Diagram when used with 5 WHY helps you in understanding the root cause of the problem. In doing so, what does the Six Sigma team wish to know?

A. The assignable causes
B. Causes attributable to categories
C. Categories that can be ignored
D. None of the above

5. Which of the below-mentioned tools is considered best to inform employees of their work roles, in terms of ease of understanding and implementation?

A. Written Instructions
B. Job Description
C. Process Flow Maps
D. Standardized Work Templates

6. In embracing Six Sigma, what has been the principal motivating factor from the point of view of upper management?

A. Bottom line results
B. Market share growth
C. Defect reductions
D. Customer focus

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7. The purpose of "rolled throughput yield" in Six Sigma would NOT be to:

A. Analyze a process flow for improvement ideas
B. Spot significant differences in yield
C. Use the calculation for customer analysis
D. Provide a baseline metric

8. A company making rubber tires is evaluated on 5 opportunities per product; in 1000 products manufactured in a month if 100 defects are observed, the Rolled Throughput Yield is:

A. 98%
B. 99%
C. 97%
D. 96%

9. What is the Lean philosophy?

A. Provide perfect value to the customer through a perfect value creation process that has zero waste
B. Continuous Improvement
C. Higher output by encouraging people to work hard and have targets
D. Reducing cost and improving purchasing power for the individuals

10. When samples are drawn out of a population randomly, what is said to be true?

A. The sample mean is always the same as the population mean
B. The sample standard deviation will be the same as population standard deviation
C. The sampling distribution approaches normality with an increase in sample size
D. The sampling distribution would be triangular if the population is distributed as triangular distribution

11. Which of the following tools is most commonly used in the define phase of a project?

A. Affinity diagram
B. Control chart
C. Failure mode and effects analysis
D. Data collection checklist

12. Which of these is considered a technique to collect data or comments from customers?

A. Multi voting
B. Customer needs prioritization
C. Focus Groups
D. Nominal Group Technique

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13. Which of the following cannot be considered as an acceptable reason for deploying a Six Sigma DMAIC project?

A. Reduce the Cost of Quality
B. Increase Customer Satisfaction
C. Reduce internal problems
D. Improve the Design

14. Sum Square of Error is equal to:

A. Sum Squares due to Pure Error + Sum Squares due to Lack of Fit
B. Sum Squares due to Pure Error - Sum Squares due to Lack of Fit
C. Sum Squares due to Lack of Fit-Sum Squares due to Pure Error
D. None of the above

15. The p-value for Moods Median test is .02 and the level of significance is set at .05, what decision is made?

A. reject null hypothesis
B. fail to reject the null hypothesis
C. cannot determine
D. use a different test

16. What type of waste would the following scenario potentially cause? Clothing manufacturing is producing more items than what the customer has ordered.

A. Waiting
B. Over Production
C. Transportation
D. Over Processing

17. Which of the following variations are identified in control charts?

A. Random and chance
B. Special and assignable
C. Normal and Special
D. Normal and random

18. Which of the following tools helps in visualizing series of causes to an effect?

A. Cause and Effect Matrix
B. Correlation Diagram
C. Ishikawa Diagram
D. Value Stream Mapping

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19. Which of the following statistical tests should be used by the Green Belt for testing the means between two inter-related groups?

A. 2 Sample t assuming equal variances
B. 2 Sample t assuming unequal variances
C. Paired t test
D. z test

20. Standard Deviation in Six Sigma applications is referred to as difference from the:

A. Target
B. Specification limits
C. Nearest fit value
D. Mean

21. In a typical DFSS Approach, which of these stages figures in most DFSS approaches?

A. Innovate
B. Identify
C. Improve
D. Control

22. Which of the following pairs is the most useful in preparing control charts, when used together, for variables data?

A. AQL, p-bar
B. p, n
C. X-bar and R
D. R, sigma

23. Which of the following activities is NOT non-value-added?

A. Setup
B. Rework
C. Process
D. Inspection

24. Repetition helps the Six Sigma team in determining:

A. Short term variability
B. Long term variability
C. All of the above
D. None of the above

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25. Identify the most accurate method for quantifying gage repeatability and reproducibility.

A. The bias and linearity method
B. The range method
C. The average and range method
D. The ANOVA method

26. Which of these tools/techniques represent "continuous improvement"?

A. Kaizen
B. Six Sigma
C. Lean
D. Kanban

27. A Six Sigma project can reduce:

A. Internal Failure Cost
B. External Failure Cost
C. Total Failure Cost
D. None of the above

28. Why would Six Sigma activities prompt a re-analysis of measuring systems?

A. It's mandated by the MSA (1998) Reference Manual
B. Variability may be reduced, requiring more precise measurements
C. Six Sigma is a more quality-conscious control technology
D. Most six sigma training includes statistical software which requires frequent re-analysis

29. Japanese 5S methodology is created and used for:

A. Continuous Improvement
B. Prevent Defects
C. Creating a productive work environment
D. Reduce Variation

30. What does OEE stands for?

A. Overall Equipment Effectiveness
B. Overall Estimation Effectiveness
C. Overall Equipment Estimation
D. Overall Effective Estimation

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31. The house of quality is used to translate customer wants into engineering design variables. The linking or prioritization of customer wants into engineered values occurs in which of the following elements?

A. Competitive analysis
B. Conflict analysis
C. Technical review
D. Relationship matrix

32. What is the name of the condition if you are running a 7-4 fractional experiment? You know factors A, B, C, D, and E are independent of each other, but you suspect factors F and G are not independent. You conducted a small sub-experiment and discovered a high correlation between factors F and G.

A. Collinearity
B. Confounded
C. Correlation
D. Covariates

33. Most of the modern computer programs can perform an analysis of experimental residuals. Which of the following techniques will NOT be employed?

A. Control Charts
B. Histograms
C. Normal probability plots
D. Dot plots

34. If an experiment has an alias, you could say that the two-factor effects are:

A. Confounded
B. Blocked
C. Misnamed
D. Mixtures

35. What types of variation does a process have if it is "in control"?

A. Process has assignable causes
B. Process has assignable and chance causes
C. Process has random causes only
D. Process has assignable or chance causes

36. For 4 factors and 3 levels, the DOE setup has 32 runs. What type of experiment is being conducted?

A. Half fractional factorial
B. Full factorial
C. Quarter fractional factorial
D. Response Surface

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37. Which of the following techniques used in DOE helps you in identifying the pure error?

A. Replication
B. Blocking
C. Randomization
D. Coding

38. A combined calculation of repeatability and reproducibility using the average and range method produces a ratio of 7.42% of process tolerance. What can be stated about the 7.42% value?

A. The measurement system is acceptable
B. The measurement system is marginal
C. The measurement system is not acceptable
D. It must be separated into individual R&R values

39. In a single factor ANOVA, the assumption of homogeneity of variances applies to:

A. The variances within the treatment groups
B. The variances of the treatment groups
C. The total variance
D. All of the above

40. A perfect regression model with all points fitting the regression line has Sum Square of Errors _____.

A. >0
B. <0
C. equal to 0
D. equal to 1

41. A QFD template allows the Six Sigma team to learn which of the following?

A. The most important customer requirements
B. The performance of a company versus competitors
C. The core area of focus for efforts
D. All of the above

42. What are the six experiments called while performing one experiment with four repetitions?

A. Randomization
B. Replications
C. Blocking
D. Sequencing

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43. If men having high Blood Sugar problems are diagnosed with Diabetes, with the mean blood sugar level to be at 120 and a standard deviation of 9, and any individual having greater than 100 Blood Sugar levels can be diagnosed with Diabetes, what is the probability of committing a Type II Error?

A. 0.007
B. 0.0062
C. 0.0139
D. 0.0055

44. A Six Sigma Green Belt practitioner constructs a control chart to display a process mean and its outer limits. In such a chart, what does UCL stand for?

A. Upper Cycle Length
B. Upper Control Limit
C. Upper Cycle Limit
D. Upper Control Length

45. To optimize the response of a process, ideally what sequence of experimentation should one use?

A. Use response surface methodologies at all stages
B. Use screening first and then response surface techniques
C. Use charting techniques first and then ANOVA
D. Use experimental designs first and then ANOVA

46. Which of the following terms describes the situation in which the effects of two factors are correlated while designing experiments?

A. Covariate
B. Confounded
C. Interactive
D. Collinear

47. What is the process capability for attribute data?

A. Cannot be determined
B. Is determined by the control limits on the applicable attribute chart
C. Is defined as the average proportion of nonconforming product
D. Is measured by counting the average nonconforming units in 25 or more samples

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48. An organization finds that an extra measure of security added to online transactions did not reduce the incidence of identity theft. What is this concept of management referred to?

A. no-value added
B. no-security added
C. limited effectiveness model
D. identity theft model

49. Which of the following statements is NOT true about control charts?

A. They can detect trends of statistical significance
B. They provide straightforward, easily interpreted information
C. They provide an ongoing measure of process capability
D. They can detect special causes of variation

50. Rolled Throughput Yield best indicates:

A. The quality levels of the process
B. The need for improvement
C. The time and effort spent on rework and scrap
D. The utilization of the process

51. For a full factorial experiment with 23 treatments and 0 replicate, how many runs could you expect?

A. 23
B. 16
C. 46
D. 32

52. What set of basic quality tools would be most applicable for a work team to use when there is a desire to follow procedures and work instructions more closely?

A. Pareto and affinity diagrams
B. Datasheets and histograms
C. Checklists and flow charts
D. Fishbone and control charts

53. Hypothesis tests on enumerative statistics would NOT involve which of the following distributions?

A. Chi-square
B. Student's t
C. Binomial
D. Poisson

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54. One of the purposes of using a fishbone diagram is to:

A. Separate a problem into smaller components
B. Identify the significant few components of a problem
C. Define the problem in sequential order
D. Show the relationship between parameters

55. Which of the following control charts would best fit a process in which measurement data on a product is easily obtained?

A. Run charts
B. Median charts
C. X-bar and R charts
D. p charts

56. With which of the following distributions is the term "degrees of freedom" inapplicable?

A. F Distribution
B. Poisson
C. Student's t
D. Chi-square

57. What determines the factors that will receive major attention after constructing a rank order prioritization matrix?

A. The relative strength values of the factors
B. Those criteria with the highest numeric values
C. Those criteria at the top of the chart
D. The factors with negative strength values

58. Which of the following is an example of a Primary metric for Six Sigma projects?

A. Quality
B. Morale of employees
C. Average turnover
D. Customer Satisfaction

59. What skills and experiences do project improvement team members normally have?

A. Narrow skills and experiences
B. Diverse skills and narrow experiences
C. Diverse experience and narrow skills
D. Diverse skills and experiences

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60. What type of variation is undesirable when trying to control a process?

A. Special cause variation
B. Random cause variation
C. Chance cause variation
D. Normal variation

61. An engineer is considering a fractional factorial instead of a full factorial to analyze a process because of a large number of variables under study. Apart from the possibility of studying a large number of factors with relatively few experiments, what other characteristics will support a decision to use a fractional factorial instead?

A. It is suspected that there are many interactions
B. The process is well known and only the main factors are of concern
C. A fractional factorial will determine the main effects curvature
D. Blocking is necessary to account for nuisance factors in this study

62. The difference between strategic quality goals and the strategic business plan is that:

Strategic quality goals are often a lower tier than the strategic business plan

A. They are determined only by top management
B. They may offer conflicting priorities
C. They are based on priorities given by all levels of the company

63. Identify the post-improvement tool which would be most beneficial when generating fresh ideas after the results of an improvement process have been disappointing.

A. A post-improvement capability analysis
B. A post-improvement brainstorming session
C. A follow-up FMEA study
D. A multi-vari re-analysis

64. Which of the following errors is typically associated with the notion, "False negative"?

A. Type I error
B. Type II error
C. Type III error
D. Depends on the experiments

65. When the data has outliers, which of the measures of central tendency should be used?

A. Mean
B. Median
C. Mode
D. None of the above

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66. What type of waste is "an accountant going to the first floor to pick up documents from the printer"?

A. Overproduction
B. Motion
C. Radiation
D. Underutilization

67. What is the best choice in the hands of a Six Sigma team to increase the power of the experiments?

A. Reduce Significance level
B. Reduce Sample Size
C. Increase Sample Size
D. All of the above

68. Which hypothesis test is similar to the 1 sample t-test but is used when the data set is not-normal?

A. Mann-Whitney
B. Kruskal-Wallis
C. 1-sample sign
D. Friedman

69. Which approach talks about equipment effectiveness?

A. Lean
B. TPM
C. Six Sigma
D. TOC

70. What is the more appropriate definition of Non-Value-Add Activity?

A. There is no monetary value involved in activity
B. Any activity that does not add any value to the product
C. Non-tangible value
D. The improvement activities cannot be valued by the organization

71. The best metric for measuring defectives is:

A. DPMO
B. DPU
C. PPM
D. DPO

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72. If the Measure stage in a DMAIC approach is all about measuring the data for an existing process, what is the main objective in the Measure phase in a DFSS approach?

A. Measure what you can measure
B. Measure what the customer wants you to measure
C. Set the key measurement steps and metrics
D. Define operational metrics

73. All the factors that might contribute to a production problem must be discovered. Which among the following problem-solving tools might be the best choice?

A. Pareto diagrams
B. Fishbone diagrams
C. Histograms
D. Control charts

74. One thousand units of a product were examined for the possibility of 5 different undesirable characteristics. A total of 80 defects were found. How many defects would be expected in a million opportunities?

A. 16000
B. 26666
C. 61458
D. 80000

75. On a highly automated food processing line, a quality professional wants to chart the weight of packages. The recommended control chart is an X-bar - S chart and not the typical X-bar - R chart, in wide use throughout the facility. Which of the following is the most logical reason for this switch?

A. The X-bar control limits will be tighter
B. The supervisor obviously wants some variety in control chart usage
C. Only one control chart will be required
D. The X-bar and S values will come automatically from a weight checker

76. Variable control chart subgroup sizes are generally 3, 4, 5, or 6 for all of the following reasons EXCEPT:

A. They are large enough so that averages of data will follow the normal distribution
B. They fit onto traditional chart paper very well
C. Larger sizes permit an opportunity for process changes within the subgroup
D. They permit separation of within time from time-to-time variation

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77. Which of the following measures of central tendency is preferred when the data does not have outliers?

A. Mean
B. Mode
C. Median
D. None of the above

78. Poke Yoke is a lean manufacturing concept that is used for what purpose?

A. Process leveling
B. Mistake proofing
C. Process visualization
D. Quick set up

79. Which of the following assumption is necessary to validate the meaning of the standard deviation of measurement variability?

A. Measurement errors are independent
B. The measurement scale is normally distributed
C. Measurement errors are independent of the operators involved in the study
D. Measurement errors are skewed in the direction of normality

80. For many people, changing a single tire can easily take 15 minutes. But, for a NASCAR pit crew, changing four tires takes less than 15 seconds. This represents a good use of which tool? SMED (Single-Minute Exchange of Die)

A. Poka Yoke
B. Kanban
C. Six Sigma

81. In a study of probability, an event or outcome that cannot be broken down any further is referred to as:

A. A mutually exclusive event
B. A dependent event
C. A simple event
D. An event complement

82. As a Green Belt, what should a trainee do when he faces a problem during his projects?

A. Quantify the problem and consult the Black Belt
B. Consult the Champion
C. Consult Team Members
D. Consult MB

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83. Which of these distributions can be used to make inferences about population variances?

A. T distribution
B. Z distribution
C. Chi-Square distribution
D. Gamma distribution

84. Which of the following is the LEAST likely tool to assist the problem definition stage of Six Sigma?

A. CTQ trees
B. Pareto analysis
C. Product yield data
D. Control charts

85. For a process working at 5 Sigma level, how many opportunities are considered to lie outside of the specification limits provided by the customer?

A. 233
B. 6210
C. 3.4
D. 66807

86. In a typical Measure Phase, which of the following activities should a Green Belt perform first?

A. Stability
B. Capability
C. MSA
D. Normality

87. In a typical MSA GAGE RR study conducted, what should the Six Sigma team determine about the Measurement System first?

A. Accuracy
B. Stability
C. Resolution
D. Linearity

88. Which of the following scenarios is best suited to use an X-bar and R chart?

A. Smaller sample size is needed
B. It is necessary to know when to investigate a process for causes of variation
C. The machine capability is wider than the specification
D. An acceptable quality level must be established

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89. If 6 consecutive samples were taken from a process and precisely measured, you can still expect differences. What type of variation would be the most difficult one to determine?

A. Lot-to-lot variation
B. Piece-to-piece variation
C. Inherent process variation
D. Error of measurement

90. What is one basic assumption of a null hypothesis?

A. That the variables are dependent
B. That the variables are independent
C. That the sample size is adequate
D. That the confidence interval is ± 2 standard deviations

91. As an experienced experimenter, you have built a predictive model of experimental data. The difference between the actual response data and the model data are termed as:

A. Confounded data
B. Nested experiments
C. Residuals
D. Efficiency of estimators

92. In the theory of control charts, the distribution of the number of defects per unit follows which of the following distributions?

A. Normal distribution
B. Binomial distribution
C. Chi-square distribution
D. Poisson distribution

93. What type of waste is "Purchasing and storing office supply, sales literature, etc."?

A. Overproduction
B. Cost
C. Inventory
D. Motion

94. For a Six Sigma project done in the Operations department (LOB project), which of the following could be called a stakeholder to the project?

A. IT Department
B. Administration Department
C. Human Resources Department
D. All of the above

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95. Which item should not be identified in the Define Phase?

A. Root causes
B. The key problem area
C. Possible financial loss
D. Intangibles

96. The distribution that follows principles of an exponential distribution is:

A. Poisson
B. Binomial
C. Chi-Square
D. Normal

97. While considering the stakeholder groups, which of the following terms is closely identified with the term 'community'?

A. Society
B. Government
C. External customers
D. Suppliers

98. Poka-yoke is best defined as:

A. capturing the voice of the customer
B. improving machine efficiency
C. reducing field failures to virtually zero
D. preventing controllable defects

99. Which of the following is NOT necessary for the Six Sigma team to update in the Project Charter in the Define Phase?

A. Project Name and Description
B. Business need
C. Project purpose
D. Constraints

100. Which of the following distributions describes the ratios of two variances drawn from the same normal population?

A. F statistic
B. Student's t test
C. Chi-square
D. Normal

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